Scalability & Future Plan

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Static CSV based data storage**	Data is stored in local files, making scaling and real-time updates difficult	High
2	**Limited model explainability**	Predictions do not explain why a crop is recommended	High
3	**Synchronous Flask inference**	Model runs in blocking mode, reducing performance under heavy traffic	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load Handling	Single Flask server running locally	Deploy using Docker + Gunicorn on AWS / Render with auto-scaling
2	Data Storage	CSV based dataset stored locally	Migrate to PostgreSQL / MySQL for structured and scalable storage
3	Performance Optimization	Direct synchronous model prediction	Use caching (Redis) + batch processing for faster inference
4	Security	No authentication or encryption	Add HTTPS, JWT authentication, and secure API endpoints

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	**Explainable AI (SHAP / LIME integration)**	2026 (1-2 Months)	Provides clear reasons for crop recommendation decisions
Phase 2	**Database Integration & REST APIs**	2026 (3-4 Months)	Enables scalable data storage and external system integration
Phase 3	**Smart Farming Integration (IoT + Sensors)**	2026(5-6 Months)	Real-time soil and climate monitoring for dynamic predictions